Concept 1 | Fundamentals of Random Variables

English Student Edition • Focused formulas and guided practice

Formula opener

Expectation

$$E(X) = \sum xp(x)$$

Variance

$$\mathrm{Var}(X) = E(X^2) - [E(X)]^2$$

Probability

$$\sum p(x) = 1$$

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Worked Solutions **Fundamentals of Random Variables**

Q001 **Written**

For the sum of two dice, construct the probability distribution and find P5 $<= X <= 8$.

Solution

1. List the 36 equally likely pairs, count each sum, and divide each frequency by 36.

Final answer

Values 2 to 12 have frequencies 1,2,3,4,5,6,5,4,3,2,1 over 36; P5 $<= X <=$ 8=20/36=5/9.

Q002 **Written**

Find the probability distributions for the absolute difference of two dice and the number of white balls drawn from the stated bag; evaluate the requested interval probabilities.

Solution

1. Check that all table probabilities add to 1 before evaluating an interval event.

Final answer

Use the sample space or combinations to list every possible X and sum the relevant probabilities.

Worked Solutions **Fundamentals of Random Variables**

Q003 **Written**

Construct both source probability tables (four coins; three balls from 4 red and 2 blue) and evaluate the given ranges.

Solution

1. A probability distribution is a table of every possible value of X and its probability.

Final answer

Use binomial or hypergeometric counts, then add probabilities for the requested values.

Q004 **Written**

Find EX for tails in four coins and for red balls in a draw of three from a bag with 4 red and 5 white.

Solution

1. Build the table, multiply each value by its probability, and add the products.

Final answer

For four fair coins EX=2; for the hypergeometric draw EX=4/3.

Worked Solutions **Fundamentals of Random Variables**

Q005 Written

Find expected values for tails in three coins, the absolute difference of two dice, and white balls in the stated bag.

Solution

1. Keep the random variable definition unchanged while forming the table.

Final answer

Use EX=sum xP$X = x$ after constructing each distribution.

Q006 Written

Find EX for all six source random-variable experiments (cards, dice, balls, tickets, coins and repeated die rolls).

Solution

1. The expected number of successes in repeated identical trials can be checked with n times p.

Final answer

Compute each mean from its probability distribution; symmetry gives quick checks for fair cards and dice.

Worked Solutions **Fundamentals of Random Variables**

Q007 Written

For tails in four fair coins, find E(3X+2), E(-4X+1), and EX^2.

Solution

1. Constants shift the mean; a multiplier scales the mean by the same factor.

Final answer

Use $E(aX + b) = aEX$+b; calculate EX^2 directly from the table.

Q008 Written

For white balls drawn from 4 red and 3 white, find the four requested transformed expected values.

Solution

1. Do not replace EX^2 by EX^2; they are different quantities.

Final answer

Apply linearity for aX+b and use the weighted square sum for X^2.

Worked Solutions **Fundamentals of Random Variables**

Q009 **Written**

Solve both source Exercise groups of transformed expectations for coins and for balls.

Solution

1. Write the transformation before substituting the known mean.

Final answer

For every expression use $E(aX + b) = aEX$+b; evaluate powers from the distribution table.

Q010 **Written**

For black balls drawn from the stated box, find EX and VX.

Solution

1. The distribution is PX = 0=2/7, PX = 1=4/7, PX = 2=1/7. Hence EX=6/7 and VX=EX^2-EX^2 = 20/49.

Final answer

$EX = 6/7; VX = 20/49.$

Worked Solutions **Fundamentals of Random Variables**

Q011 Written

Find the mean and variance for the two source Try random variables (a numbered card and winning tickets).

Solution

1. Variance measures spread around the mean, not the mean itself.

Final answer

Construct each probability table, then use VX=EX^2-EX^2.

Q012 Written

Find EX and VX for all five source Exercise random variables.

Solution

1. For sampling without replacement, use combinations to obtain the exact probabilities.

Final answer

Use the distribution table for each experiment and the variance identity.

Worked Solutions **Fundamentals of Random Variables**

Q013 **Written**

Find the standard deviation of tails when two biased coins have P(head)=4/7 and P(tail)=3/7.

Solution

1. Find VX first, then take the positive square root.

Final answer

The table gives VX=24/49, so sigma$X$=2 sqrt6/7.

Q014 **Written**

Find the standard deviation from each source probability table and for white balls drawn from 4 red and 3 white.

Solution

1. Standard deviation has the same units as X and is always non-negative.

Final answer

Use sigma$X$=sqrt$V(X)$ after evaluating the variance from the table.

Worked Solutions **Fundamentals of Random Variables**

Q015 Written

Find the standard deviation for both source probability-table exercises.

Solution

1. Keep the positive square root only.

Final answer

Compute EX, EX^2, then sigma$X$=sqrt$E(X^2) - E(X)^2$.

Q016 Written

For the number of winning tickets in the source draw, find VY and sigma$Y$ for Y=2X+1 and Y=-3X-2.

Solution

1. The constant b changes neither variance nor standard deviation.

Final answer

Use $V(aX + b) = a^2VX$ and sigma(aX+b)=|a|sigma$X$.

Worked Solutions **Fundamentals of Random Variables**

Q017 **Written**

For red balls drawn from the stated bag, find variance and standard deviation for the three linear transformations of X.

Solution

1. Write both transformation rules before substituting numbers.

Final answer

Apply the squared multiplier to VX and the absolute multiplier to sigma$X$.

Q018 **Written**

For a single die, find VY and sigma$Y$ for Y=X+2, Y=3X+1 and Y=-2X+4.

Solution

1. A translation changes the mean but leaves spread unchanged.

Final answer

Use the variance and standard-deviation transformation formulas for each Y.

Worked Solutions **Fundamentals of Random Variables**

Q019 **Written**

Solve the two source transformation problems: a net prize after an entry fee, and finding a,b from E(aX+b)=7/3 and sigma(aX+b)=3.

Solution

1. The mean equation determines b after the standard-deviation equation fixes a.

Final answer

Use $E(aX + b) = aEX$+b and sigma(aX+b)=|a|sigma$X$, with a>0.

Q020 **Written**

Solve the two source Try problems involving net winnings and determining a,b for a die transformation.

Solution

1. Entry fees are negative constants; they shift the mean only.

Final answer

Model the payout as $Y = aX + b$, then apply the mean and standard-deviation rules.

Worked Solutions **Fundamentals of Random Variables**

Q021 Written

Solve all four source transformation exercises, including the number-line coin walk, games, and the calibrated die model.

Solution

1. Translate the words into aX+b before calculating.

Final answer

Express every quantity as $Y = aX + b$ and use the appropriate mean, variance or standard-deviation rule.

Q022 MCQ

A probability distribution must have probabilities that sum to:

Solution

1. Check the table total.

Correct option: B

Final answer

1

Worked Solutions **Fundamentals of Random Variables**

Q023 MCQ

For two fair dice, P$X = 2$ is:

Solution

1. Only the pair (1,1) gives sum 2.

Correct option: B

Final answer

1/36

Q024 MCQ

The number of heads in three coins can be:

Solution

1. Count heads in the sample space.

Correct option: B

Final answer

0, 1, 2, 3

Worked Solutions **Fundamentals of Random Variables**

Q025 MCQ

The expected value is calculated as:

Solution

1. Multiply each value by its probability.

Correct option: B

Final answer

sum x$PX = x$

Q026 MCQ

The mean number of tails in four fair coins is:

Solution

1. Use symmetry or the table.

Correct option: B

Final answer

2

Worked Solutions **Fundamentals of Random Variables**

Q027 MCQ

For a fair die, EX is:

Solution

1. The values 1 to 6 average to 3.5.

Correct option: B

Final answer

3.5

Q028 MCQ

E(aX+b) equals:

Solution

1. Use linearity of expectation.

Correct option: B

Final answer

$aEX + b$

Worked Solutions **Fundamentals of Random Variables**

Q029 MCQ

In general, EX^2 is:

Solution

1. Square values before weighting.

Correct option: C

Final answer

weighted square sum

Q030 MCQ

If EX=2, then E(3X+2) is:

Solution

1. Calculate 32+2.

Correct option: C

Final answer

8

Worked Solutions **Fundamentals of Random Variables**

Q031 MCQ

The variance identity is:

Solution

1. Subtract the squared mean.

Correct option: B

Final answer

$EX^2 - EX^2$

Q032 MCQ

Variance can never be:

Solution

1. It is an expected square.

Correct option: C

Final answer

negative

Worked Solutions **Fundamentals of Random Variables**

Q033 MCQ

The first step in a variance problem is to:

Solution

1. The table supplies $\mathrm{E}X$ and $\mathrm{E}X^2$.

Correct option: A

Final answer

draw the distribution

Q034 MCQ

Standard deviation is:

Solution

1. Take the positive square root.

Correct option: B

Final answer

sqrt$V(X)$

Worked Solutions **Fundamentals of Random Variables**

Q035 MCQ

Standard deviation has units:

Solution

1. The square root restores the units.

Correct option: B

Final answer

same units as X

Q036 MCQ

V(aX+b) equals:

Solution

1. Constants do not affect spread.

Correct option: B

Final answer

a squared VX

Worked Solutions **Fundamentals of Random Variables**

Q037 MCQ

sigma(aX+b) equals:

Solution

1. Standard deviation is non-negative.

Correct option: B

Final answer

|a| sigma$X$

Q038 MCQ

For Y=X+3, the variance is:

Solution

1. A translation leaves variance unchanged.

Correct option: B

Final answer

VX

Worked Solutions **Fundamentals of Random Variables**

Q039 MCQ

A net prize after a fixed fee is modeled as:

Solution

1. The fee is the constant b.

Correct option: B

Final answer

$aX + b$

Q040 MCQ

If a>0 and sigma(aX+b)=3, then a is:

Solution

1. Solve a sigma$X$=3.

Correct option: B

Final answer

3/sigma$X$

Worked Solutions **Fundamentals of Random Variables**

Q041 MCQ

Changing b in aX+b changes:

Solution

1. b is a translation.

Correct option: C

Final answer

the mean only

Q042 MCQ

If EX=2 and EX^2=7, then VX is:

Solution

1. Calculate $7 - 2^2$.

Correct option: A

Final answer

3

Worked Solutions **Fundamentals of Random Variables**

Quiz WRITTEN 1 Concept Quiz · Written

Find EX and VX for the number of heads in three fair coins.

Solution

1. Use the binomial distribution.

Final answer

EX=3/2 and VX=3/4

Quiz WRITTEN 2 Concept Quiz · Written

For Y=4X-1 with VX=2, find VY and sigma$Y$.

Solution

1. Use $V(aX+b) = a^2VX$.

Final answer

VY=32 and sigma$Y$=4 sqrt2

Worked Solutions **Fundamentals of Random Variables**

Quiz WRITTEN 3 **Concept Quiz · Written**

For a fair die, find E(2X+5).

Solution

1. EX=7/2, then use linearity.

Final answer

12

Quiz WRITTEN 4 **Concept Quiz · Written**

If sigma$X$=2 and EX=3, find sigma(-5X+7).

Solution

1. Multiply standard deviation by |a|.

Final answer

10

Worked Solutions **Fundamentals of Random Variables**

Quiz MCQ 5 **Concept Quiz · MCQ**

If VX=5, V(3X+1) is:

Solution

1. Square the multiplier.

Correct option: D

Final answer

45

Quiz MCQ 6 **Concept Quiz · MCQ**

For a fair die EX is:

Solution

1. Average 1 through 6.

Correct option: C

Final answer

3.5

Worked Solutions **Fundamentals of Random Variables**

Quiz MCQ 7 **Concept Quiz · MCQ**

A probability table total must be:

Solution

1. Add all probabilities.

Correct option: B

Final answer

1

Quiz MCQ 8 **Concept Quiz · MCQ**

If sigma$X$=4, sigma(X-9) is:

Solution

1. Translations do not change spread.

Correct option: B

Final answer

4

Answer key

Use the question number shown on each page.

Q001 Values 2 to 12 have frequencies 1,2,3,4,5,6,5,4,3,2,1 over 36; P5 <= X <= 8=20/36=5/9.

Q002 Use the sample space or combinations to list every possible X and sum the relevant probabilities.

Q003 Use binomial or hypergeometric counts, then add probabilities for the requested values.

Q004 For four fair coins EX=2; for the hypergeometric draw EX=4/3.

Q005 Use EX=sum x$PX = x$ after constructing each distribution.

Q006 Compute each mean from its probability distribution; symmetry gives quick checks for fair cards and dice.

Q007 Use $E(aX + b) = aEX$+b; calculate EX^2 directly from the table.

Q008 Apply linearity for aX+b and use the weighted square sum for X^2.

Q009 For every expression use $E(aX + b) = aEX$+b; evaluate powers from the distribution table.

Q010 $EX = 6/7; VX = 20/49$.

Q011 Construct each probability table, then use VX=EX^2-EX^2.

Q012 Use the distribution table for each experiment and the variance identity.

Q013 The table gives VX=24/49, so sigma$X$=2 sqrt6/7.

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Q019 Use $E(aX + b) = aEX$+b and sigma(aX+b)=|a|sigma$X$, with a>0.

Q020 Model the payout as $Y = aX + b$, then apply the mean and standard-deviation rules.

Q021 Express every quantity as $Y = aX + b$ and use the appropriate mean, variance or standard-deviation rule.

Q022 B

Q023 B

Q024 B

Q025 B

Q026 B

Q027 B

Q028 B

Q029 C

Q030 C

Q031 B

Q032 C

Q033 A

Q034 B

Q035 B

Q036 B

Q037 B

Q038 B

Q039 B

Q040 B

Q041 C

Q042 A

Quiz WRITTEN 1 EX=3/2 and VX=3/4

Quiz WRITTEN 2 VY=32 and sigma$Y$=4 sqrt2

Quiz WRITTEN 3 12

Quiz WRITTEN 4 10

Quiz MCQ 5 D

Quiz MCQ 6 C

Quiz MCQ 7 B

Quiz MCQ 8 B